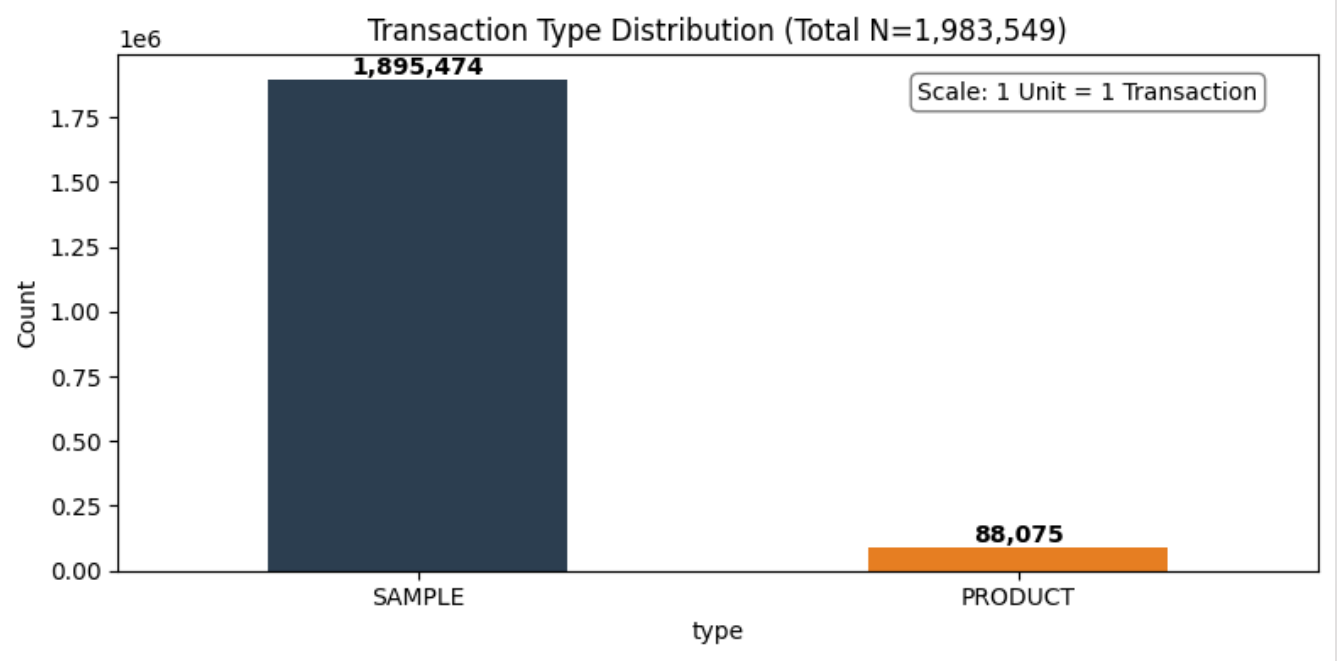
LuxeIntelligence

End-to-End Analytics & AI Ecosystem

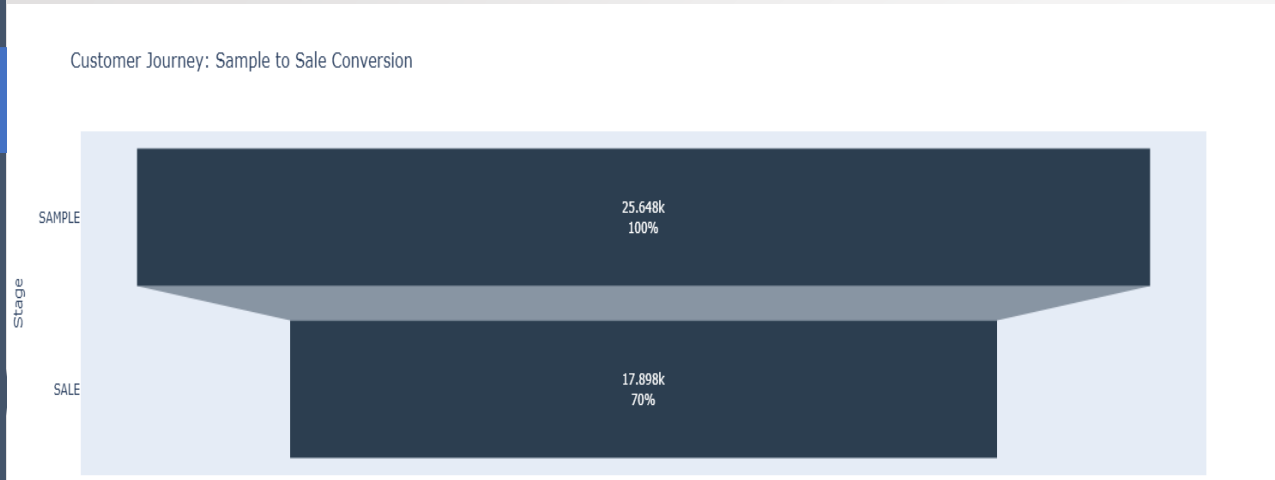
Objective & Data Foundation (Data Audit)

Objective: To transform high-volume transactional and behavioural data into a **360 Designer Intelligence Suite**. By bridging the gap between sample intent and purchase reality, this project aims to optimize conversion velocity, personalize the luxury customer journey, and provide the sales team with predictive tools to drive measurable revenue growth.

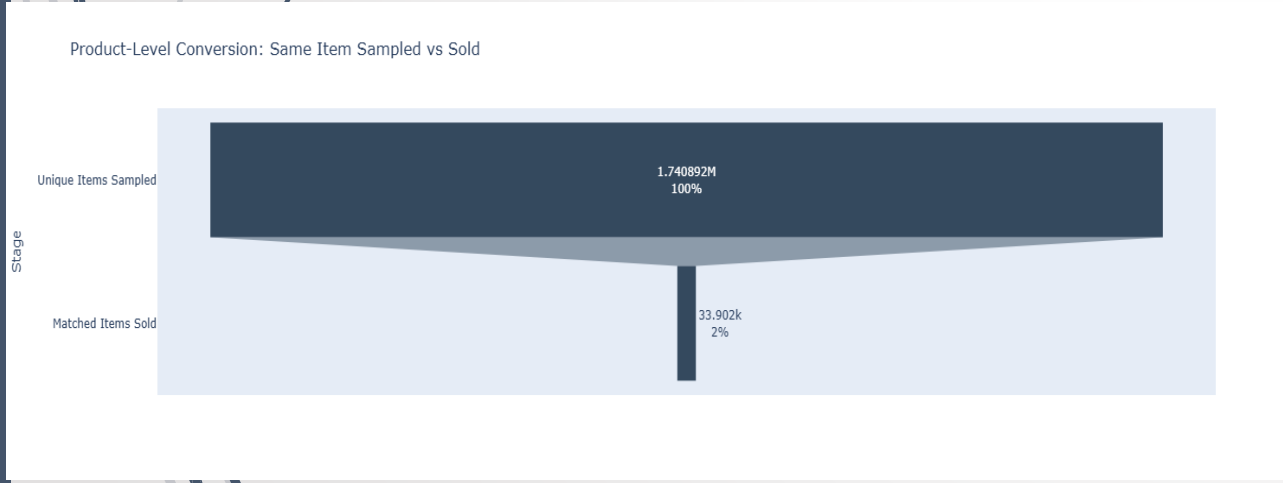
Table	Row Count	Column Count
Transactions	1,983,549	9
Item_master	16,064	10
Customer_master	29,691	4

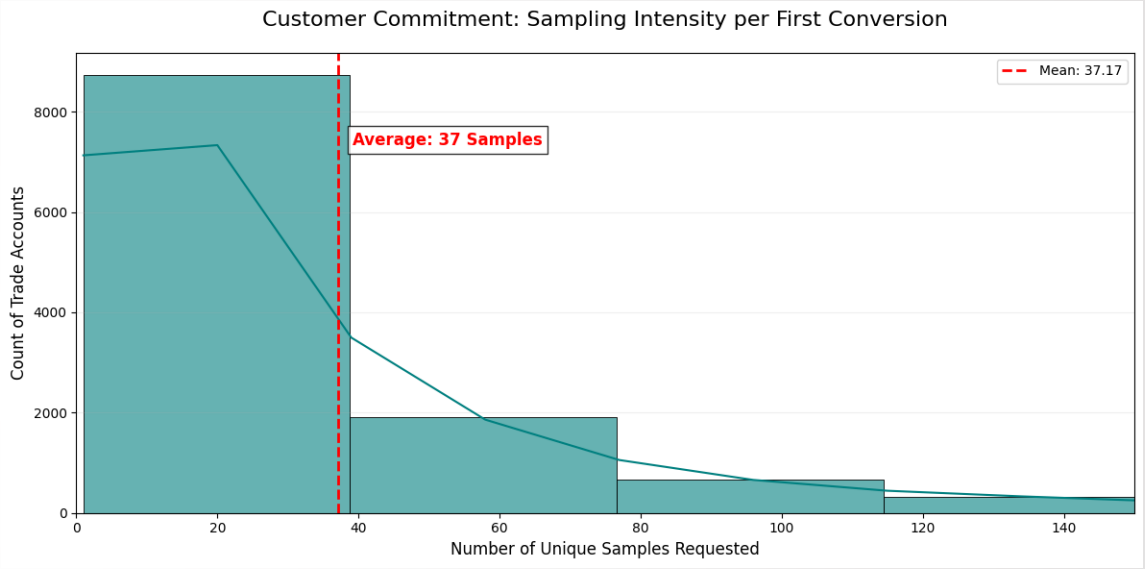


Conversion Funnel Analysis

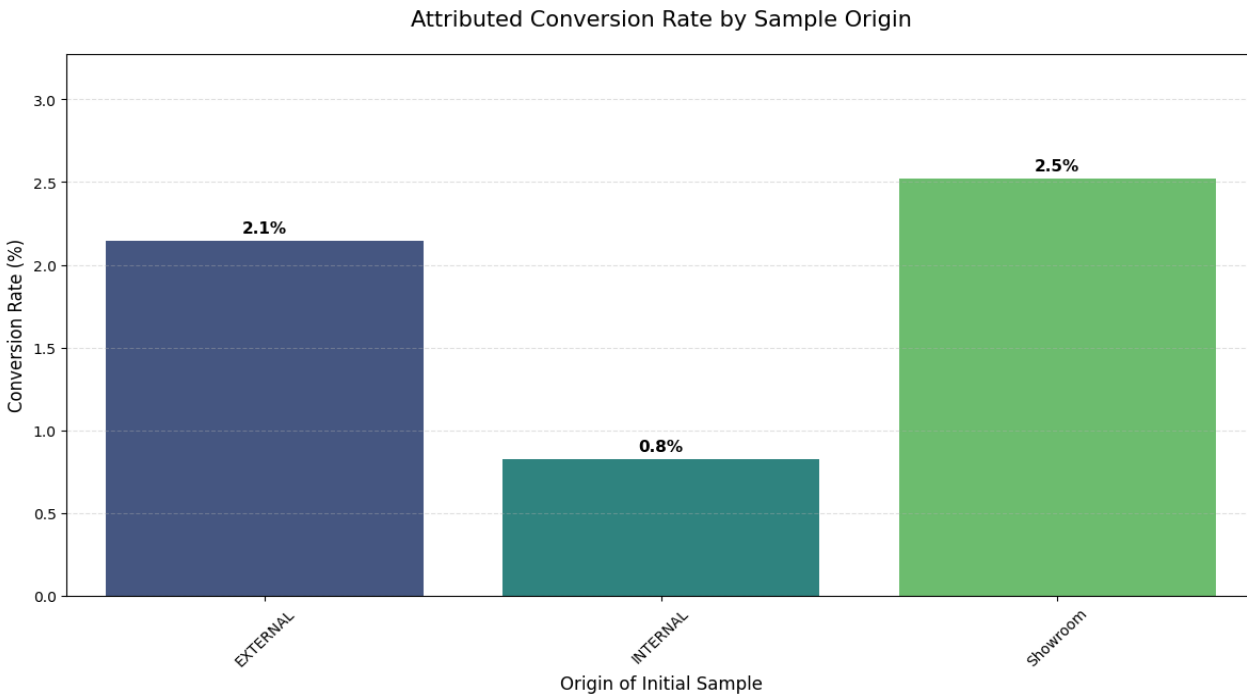


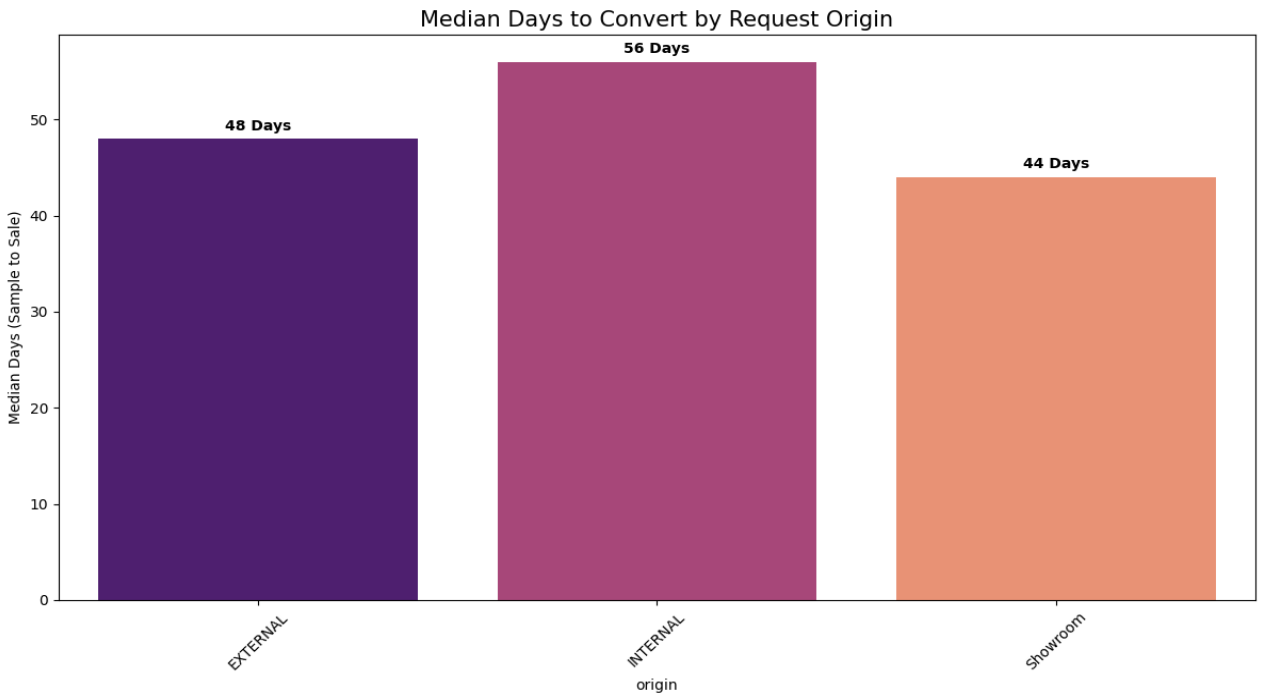
"Halo Effect" : a sample request is a high-intent signal that a design project is active. Even if the specific sampled SKU is not selected (conversion at the item level), the account remains highly likely to purchase an alternative product from the brand (conversion at the account level).





On average, a new account requires approximately **37 samples** to convert into the successful transaction.





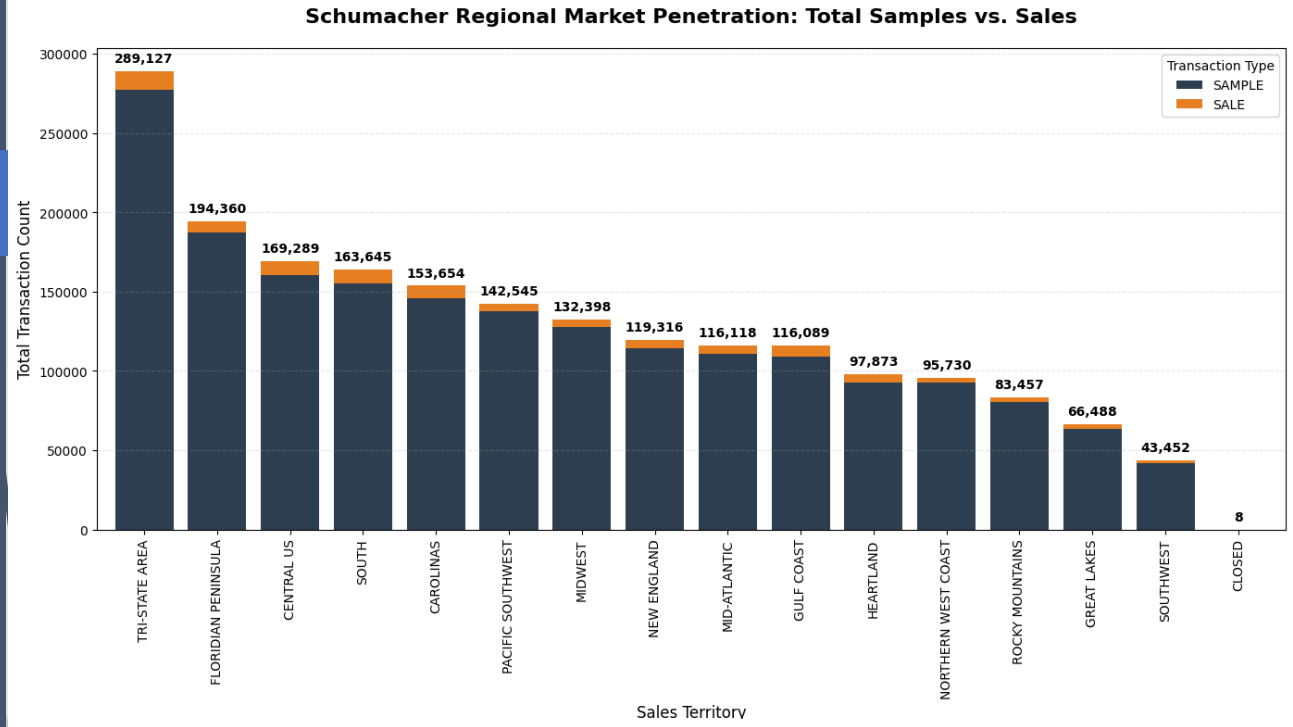
Channel Efficiency & Sales Velocity

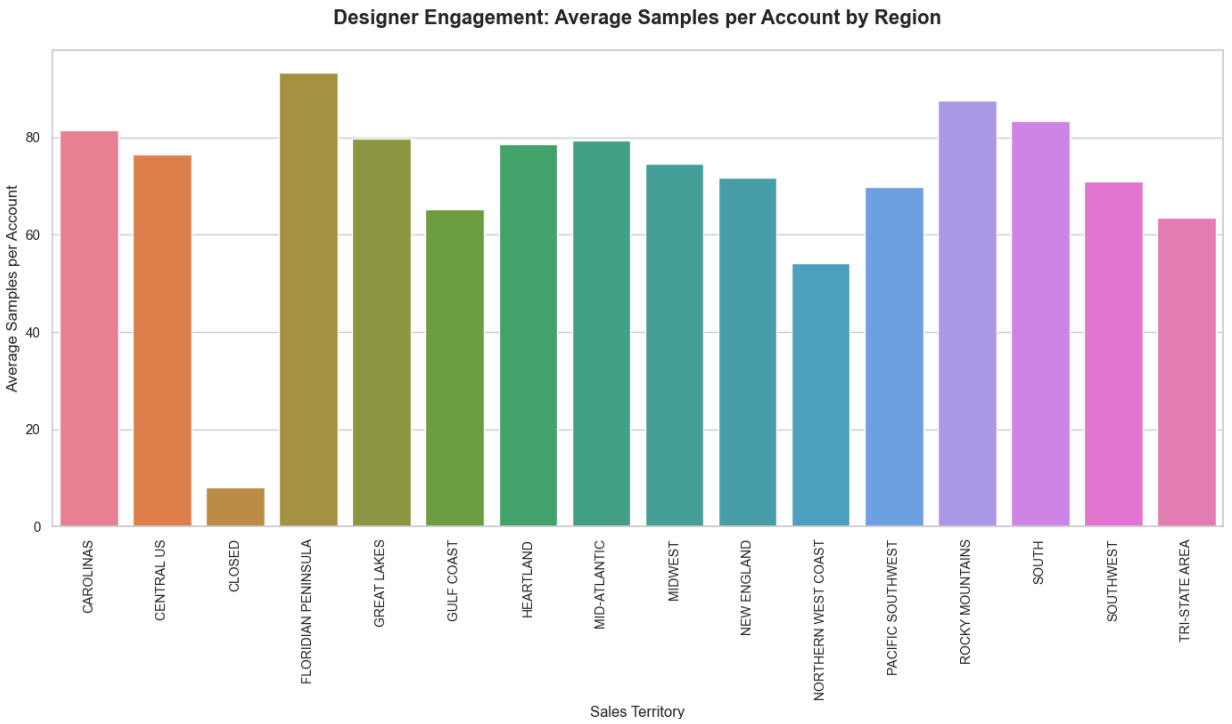
Key Finding: There is a strong correlation between **Human Interaction** and both **Speed** and **Probability** of sale.

Showroom Strength: The Showroom is our most efficient channel, converting nearly **1 in 5 leads** in just **44 days**.

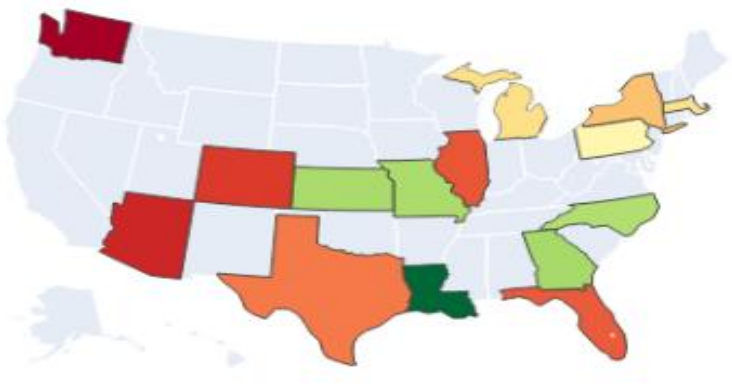
The Web Gap: Web leads have the lowest conversion (13.6%) and take nearly a week longer to close than showroom leads. This represents a significant opportunity to implement digital "White Glove" services to close the gap.

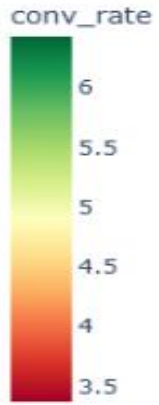
Customer Commitment: On average, a successful sale is preceded by **2.5 unique samples**, suggesting that our customers have a high degree of focus once they start the sampling process.





Regional Conversion Velocity: Sample to Sale Efficiency

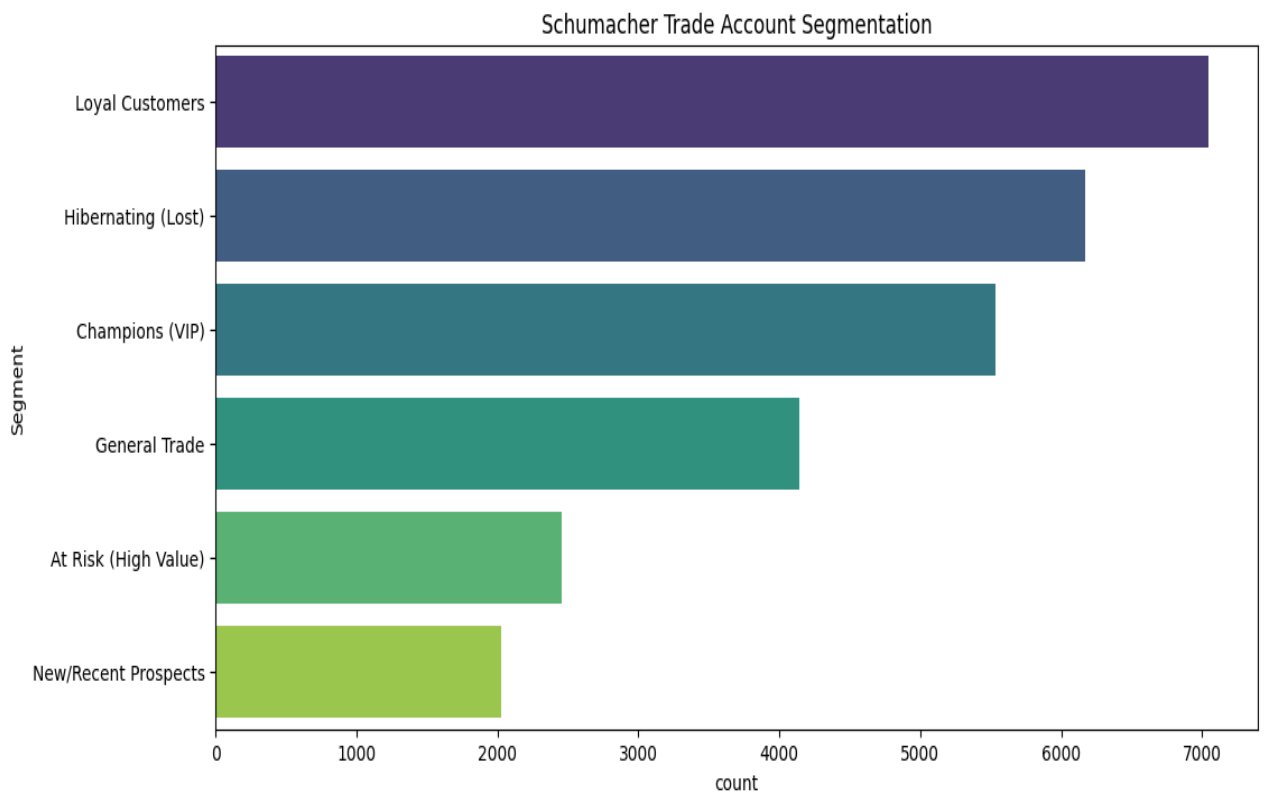




REGIONAL ANALYSIS

RFM Analysis & Segmentation

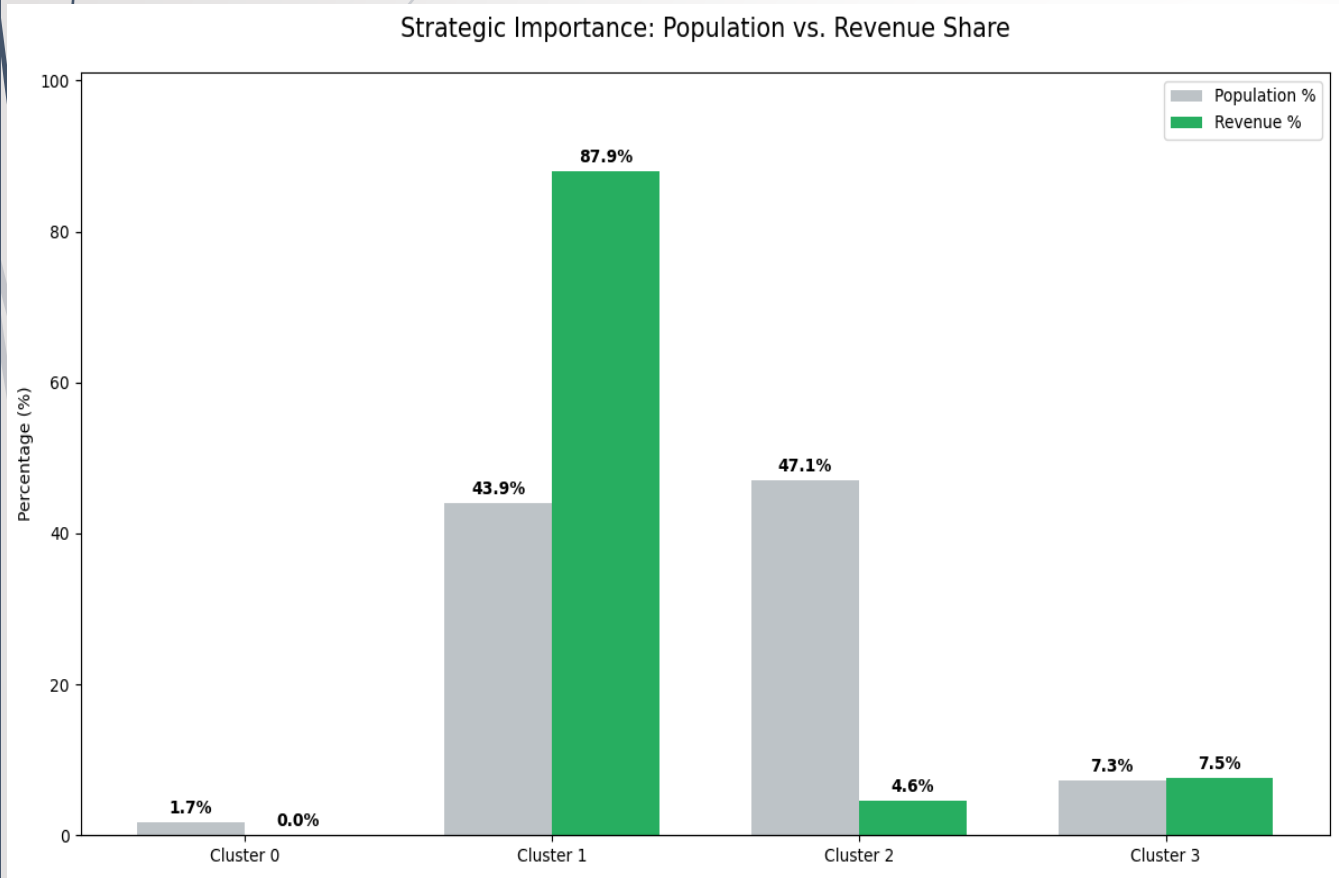
- **Recency (R):** Days since the last order.
- **Frequency (F):** Total number of unique orders placed.
- **Monetary (M):** Total net revenue generated.



Segment	Criteria (R, F, M Scores)	Business Description
Champions (VIP)	R >= 4, F >= 4, M >= 4	Your most valuable, active, and loyal designers.
Loyal Customers	R >= 3, F >= 3	Consistent purchasers who keep the business steady.
New Prospects	R >= 4, F >= 2	Recently joined or made their first small purchase.
At Risk (High Value)	R >= 2, M >= 4	Former big spenders who haven't ordered recently.
Hibernating	R >= 2, F >= 2	Inactive accounts with low lifetime value.
General Trade	All Others	The standard "middle" of the customer base.

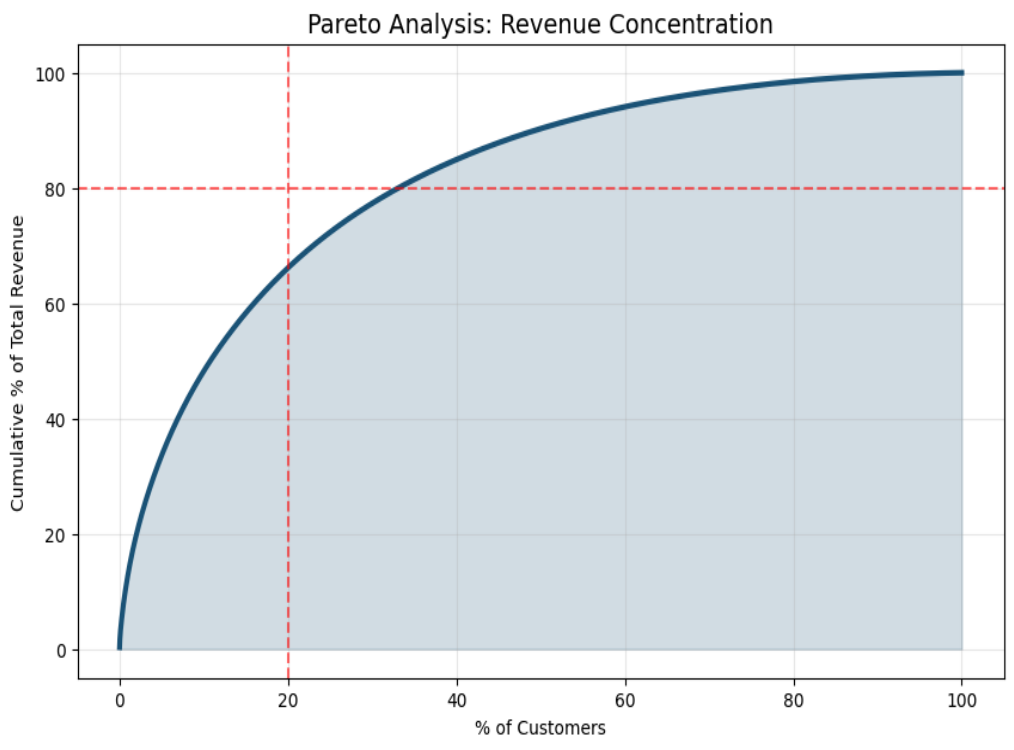
KNN IMPLEMENTATION

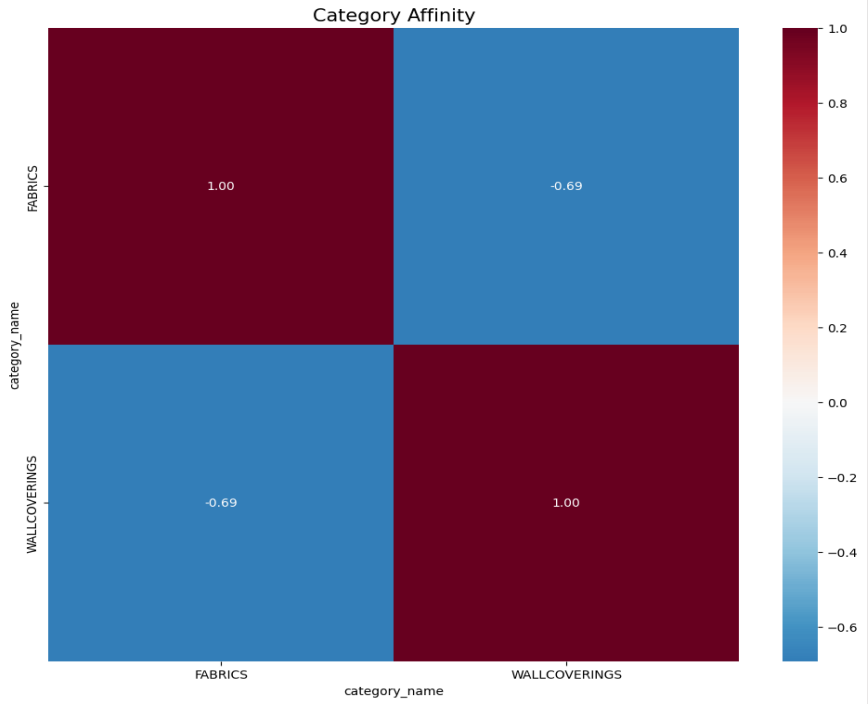
```
features = ['Recency', 'Frequency', 'Monetary']
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
rfm['Cluster'] = kmeans.fit_transform(rfm_scaled).argmax(axis=1)
```



- **Cluster 1 (Premier Trade):** Exhibits the highest median Frequency and Monetary values. The tight interquartile range (IQR) in Recency indicates a consistent, high-velocity purchasing pattern.
- **Cluster 2 (High-Intent Samplers):** Displays a disproportionately high Sample_Count relative to current Monetary value. This represents a prime conversion opportunity for the sales team.
- **Cluster 3 (Dormant/Low-Value):** Characterized by a significantly higher median Recency and lower Monetary density, indicating churn risk or one-off residential purchasers.

PRODUCT AFFINITY AND PRICING

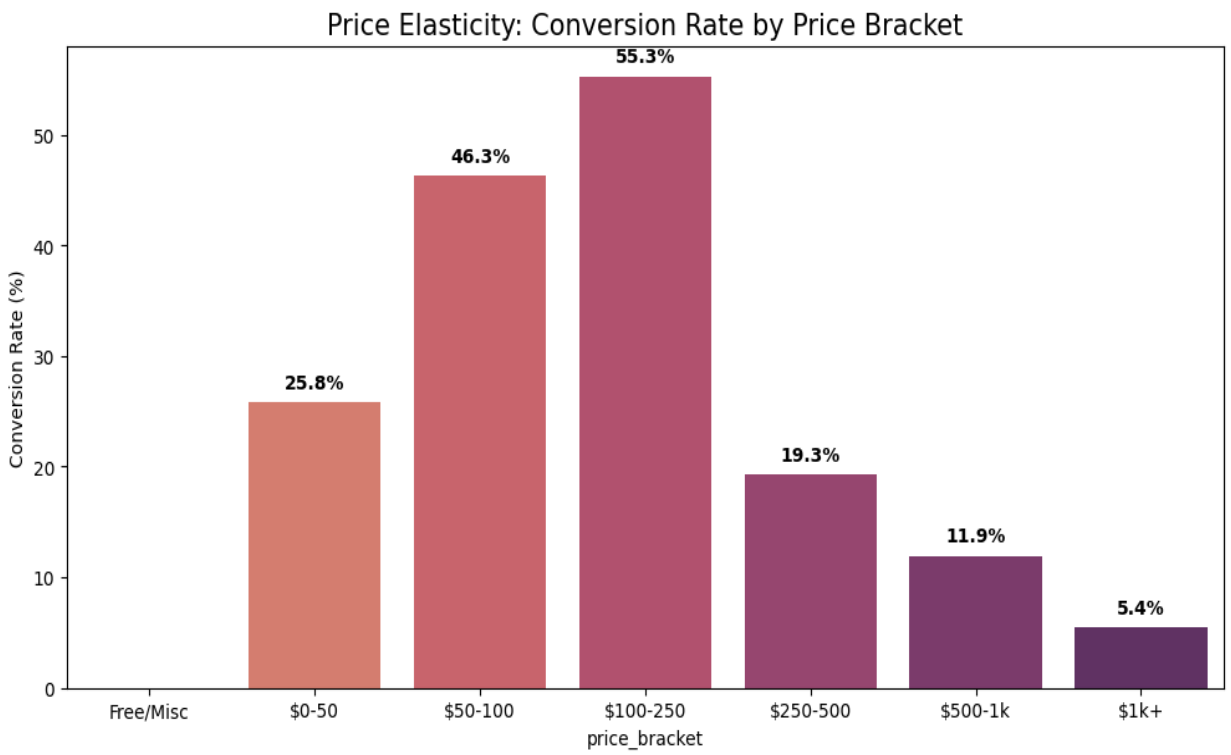


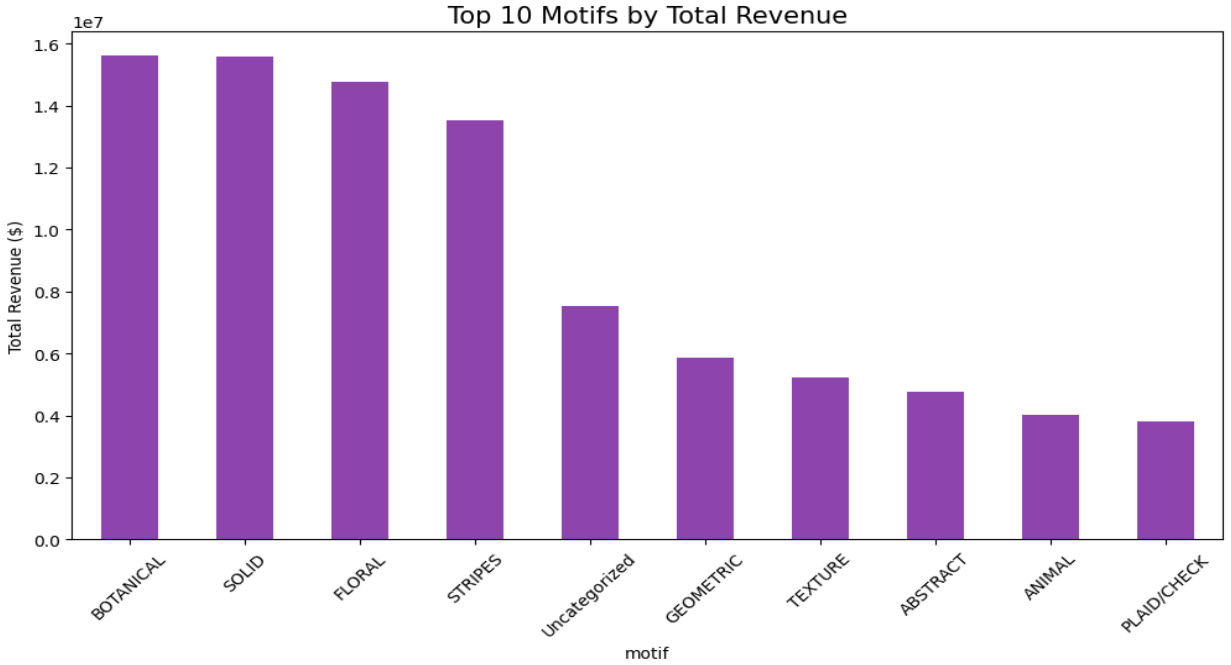


Approximately **20% of customers** contribute to roughly **80% of the total net revenue**.

Distinct Segmentation: Strong negative affinity (-0.69) suggests "Fabric-first" vs. "Wallcovering-first" buyer profiles rather than a shared basket.

Strategic Opportunity: High resistance to traditional cross-selling; requires bundled promotions or targeted "bridge" campaigns to capture total wallet share.





Goldilocks Zone ($100–250)

Positive Momentum ($0–100)

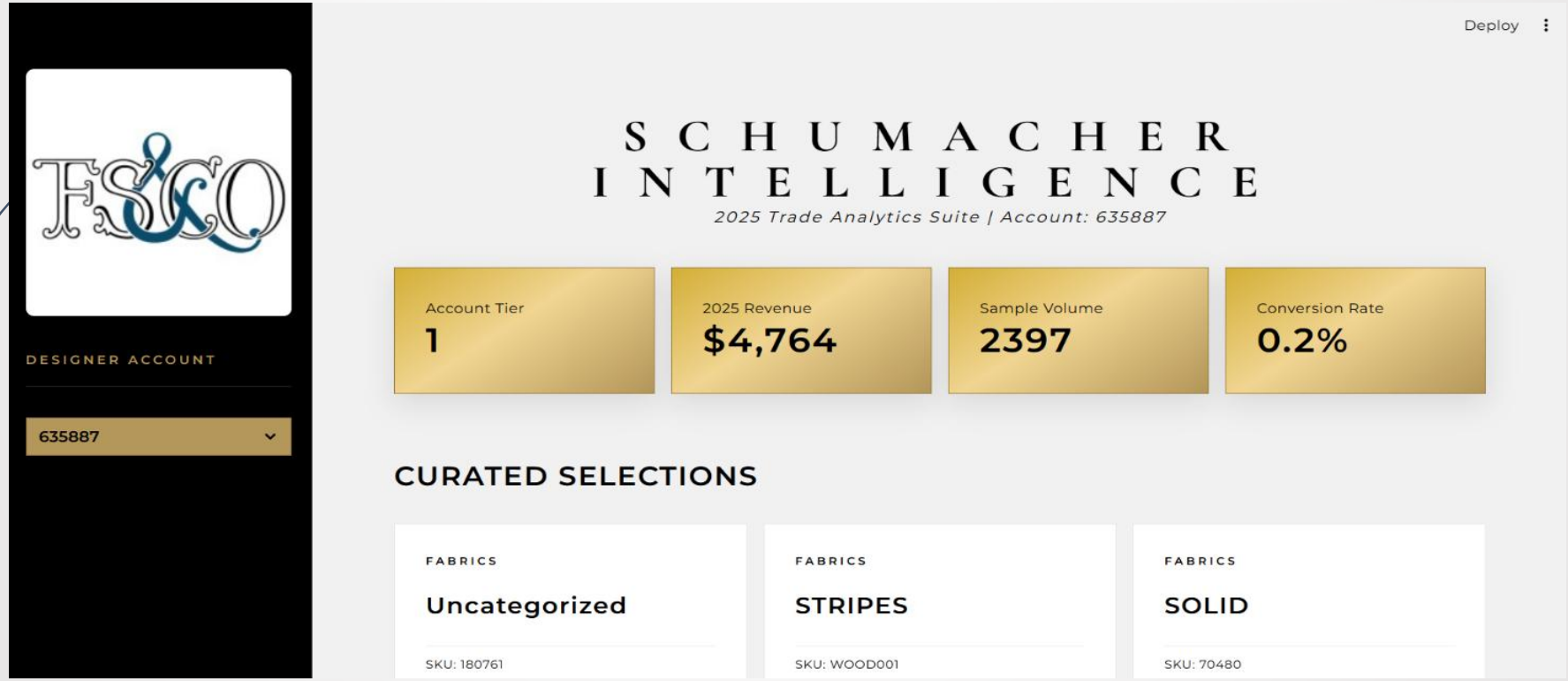
The Resistance Threshold ($250+)

High-Ticket Friction ($1k+)

The revenue analysis reveals a clear market preference for classic patterns, with **Botanical**, **Solid**, and **Floral** motifs emerging as the primary revenue drivers, each generating nearly double the individual revenue of mid-tier categories like **Geometric** or **Texture**.

Recommendation engine (AI – NLP, CV/TORCH)

Our Streamlit powered recommendation engine integrates Vision and Text AI via Hugging Face embeddings to deliver hyper-personalized curated selections, directly converting high-intent sampling into measurable revenue growth.



Recommendations

- **Precision Sales & Lifecycle Management**

Predictive Lead Scoring: Deploy a **Propensity-to-Buy Model** that analyzes request origin and historical account velocity to assign a "Lead Score" to every sample.

Automated CRM Alerts: Trigger high-priority notifications for leads with a **score > 0.8**, targeting a **10-15% increase** in conversion efficiency.

Dynamic Retention: Use **K-Means clustering** to identify "At-Risk" accounts, triggering automated API calls for personalized inspiration outreach.

- **Affinity-Based Intelligence**

Cross-Category Penetration: Utilize the **Apriori algorithm** to map statistical co-occurrences between motifs and subclasses, such as wallpaper and fabric trimmings.

"Complete the Look": Implement automated recommendations on digital Product Detail Pages (PDP) to drive **Average Order Value (AOV)**.

- **Regional ROI & Future-State AI**

Resource Allocation: Redirect marketing spend to **"High-Velocity" regions** (e.g., Gulf Coast) identified via geospatial analysis and real-time Power BI monitoring.

Vision Matcher & Forecasting: Scale the **AI engine** to allow designers to find motifs via room photos and use time-series analysis to predict demand surges for color families.

THANK YOU FOR YOUR ATTENTION